BANKWISE · MARKETING INTELLIGENCE

# Bank Marketing Prediction

Understand customer behaviour. Predict term-deposit subscription using machine learning.

Enter customer details below to predict their likelihood of subscribing to a bank term deposit.

## Model Performance

Evaluation on the held-out test set. Precision, recall and F1-score refer to the YES class.

| Model | Accuracy | Precision | Recall |
|---|---|---|---|
| Balanced Logistic Regression | 75.57% | 26.69% | 62.29 |
| Decision Tree | 84.35% | 32.02% | 30.06 |
| Random Forest | 88.00% | 48.43% | 39.32 |
| Balanced Random Forest | 84.16% | 38.37% | 58.32 |

Best F1-score: Balanced Random Forest (46.29%). Highest accuracy: Random Forest (88.00%). Highest recall: Balanced Logistic Regression (62.29%).

## Customer Information

Age: 35

Job: technician

Marital Status: married

Education: tertiary

Account Balance (€): 50000

Credit in Default: no

Housing Loan: yes

Personal Loan: no

Contact Method: cellular

Last Contact Day: 10

Last Contact Month: mar

Contacts in Current Campaign: 2

Days Since Previous Campaign Contact (-1 if none): -1

Previous Campaign Contacts: 0

Previous Campaign Outcome

failure

Predict Subscription ↻ Reset

## Prediction Results

These are model predictions, not guarantees of a customer's actual behaviour.

**Balanced Logistic Regression**

Likely to subscribe: YES

**Decision Tree**

Likely to subscribe: NO

**Random Forest**

Likely to subscribe: NO

**Balanced Random Forest**

Likely to subscribe: YES